Internal Capital Frictions in Intermediaries and Exchange Rate Dynamics

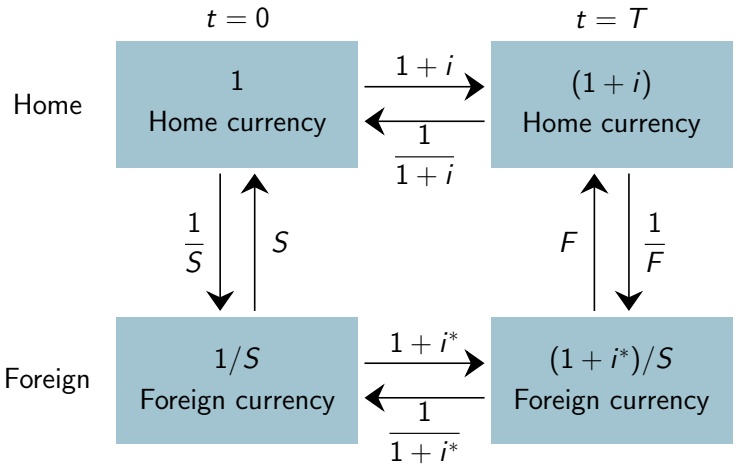
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November 4, 2026

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The Covered Interest Parity (CIP) - A No Arbitrage Condition



Pricing an FX swap:

$$\underbrace{1 + i_{0,T}}_{\text{direct}} = \underbrace{\left(1 + i^*_{0,T}\right) \frac{F_{0,T}}{S_0}}_{\text{synthetic}}$$

$$\Leftrightarrow$$

$$F_{0,T} = S_0 \frac{1 + i_{0,T}}{1 + i^*_{0,T}}$$

where i is the interest rate in Home, i^* is the interest rate in Foreign, F is the forward rate Foreign/Home, S is the spot rate Foreign/Home.

The Covered Interest Parity (CIP) - Theory vs Empirics

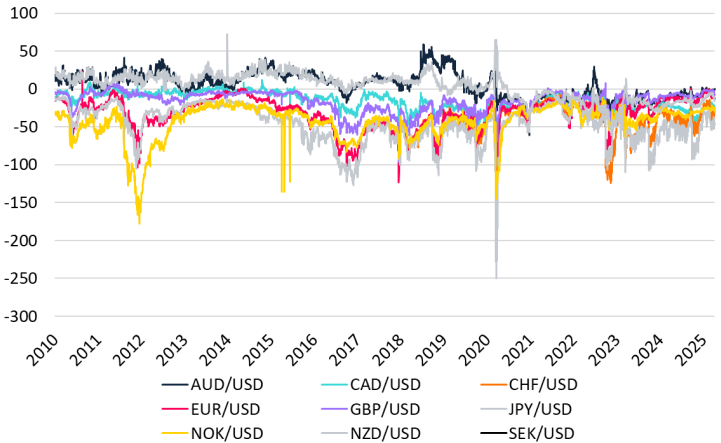
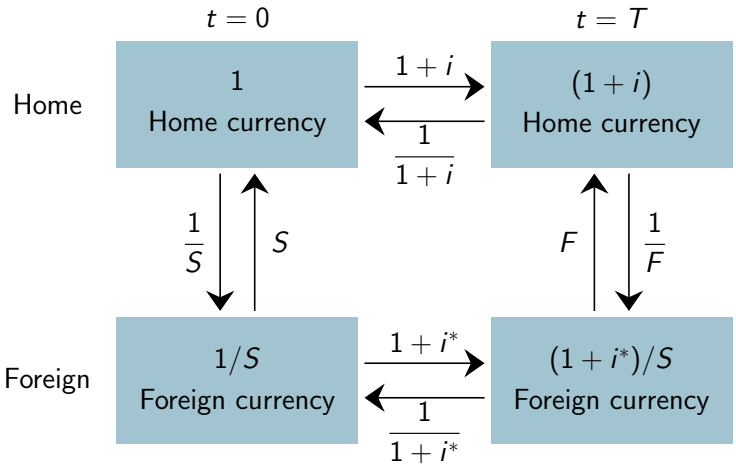


Figure: 3-month OIS-implied covered interest parity deviations for G10/USD currency pairs (bps). Sources: Bloomberg, Authors' calculations as in Siriwardane et al. (2023).

The Covered Interest Parity (CIP) - A No Arbitrage Condition?



Pricing an FX swap:

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$$\Leftrightarrow$$

$$F_{0,T} = S_0 \frac{1 + i_{0,T} - b_T}{1 + i^*_{0,T}}$$

where i is the interest rate in Home, i^* is the interest rate in Foreign, F is the forward rate Foreign/Home, S is the spot rate Foreign/Home, b_T is the cross-currency basis.

This project in a nutshell

- **Puzzle:** Intermediary asset pricing prices assets off dealer constraints...
  But the leverage ratio is *non-risk-based* and *equalizes* CIP bases b^c across currencies!
  What intermediary SDF prices the cross-section?

- **This paper:** The intermediary SDF Λ is the shadow price of *internal liquidity*.
  Variation margins M_1^c on FX forwards are uncertain and require cash, but internal funds are valuable (pecking order), so dealers demand a basis to bear VM liquidity risk.

- **Prediction:** $b^c = \mathbb{E}[\Lambda M_1^c]$, it's about the liquidity shortfall.

- **Data + identification:** Contract-level (UK-)EMIR FX-forward positions, daily valuation.
  Dealer-specific P&L shocks map where internal capital constraints bind.

- **Stylised facts:** Granular dealers, large P&L changes on FX forwards, large price variations.

Intermediary Asset Pricing, Limits to Arbitrage, FX Intermediation...

Constraints on intermediaries can move asset prices away from fundamentals

Bernanke and Gertler (1989); Kiyotaki and Moore (1997); He and Krishnamurthy (2012); Brunnermeier and Sannikov (2014); Adrian et al. (2014); He et al. (2017); Gabaix and Maggiori (2015); Itskhoki and Mukhin (2021)...

- We use CIP deviations to proxy intermediaries' SDF through internal liquidity shocks

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CIP violations reflect limits to arbitrage in segmented funding markets

Du et al. (2018); Cenedese et al. (2021); Sushko et al. (2016); Rime et al. (2022); Anderson et al. (2021); Augustin et al. (2024)... Also Shleifer and Vishny (1997), Gromb and Vayanos (2002)... Also Lustig, Roussanov, and Verdelhan (2011)...

- We isolate variation-margin shocks as convergence risk, to explain the CIP cross-section

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FX forwards remain over-the-counter and intermediated by granular dealers

Weill (2020); Hugonnier et al. (2025); Collin-Dufresne et al. (2019); Hau et al. (2021); Eisfeldt et al. (2022)...

- We document the dealer-client network, where dealer-specific shocks matter

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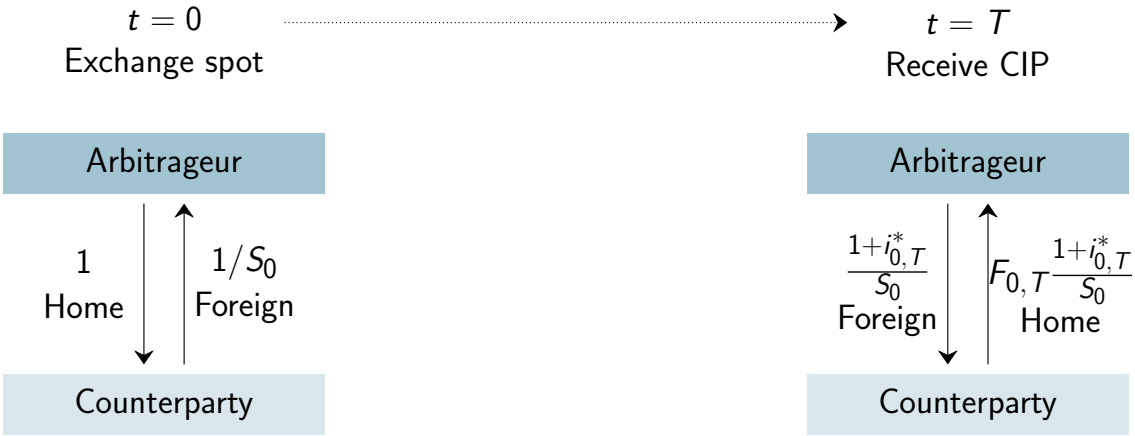
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Idiosyncratic intermediary shocks can affect aggregate prices

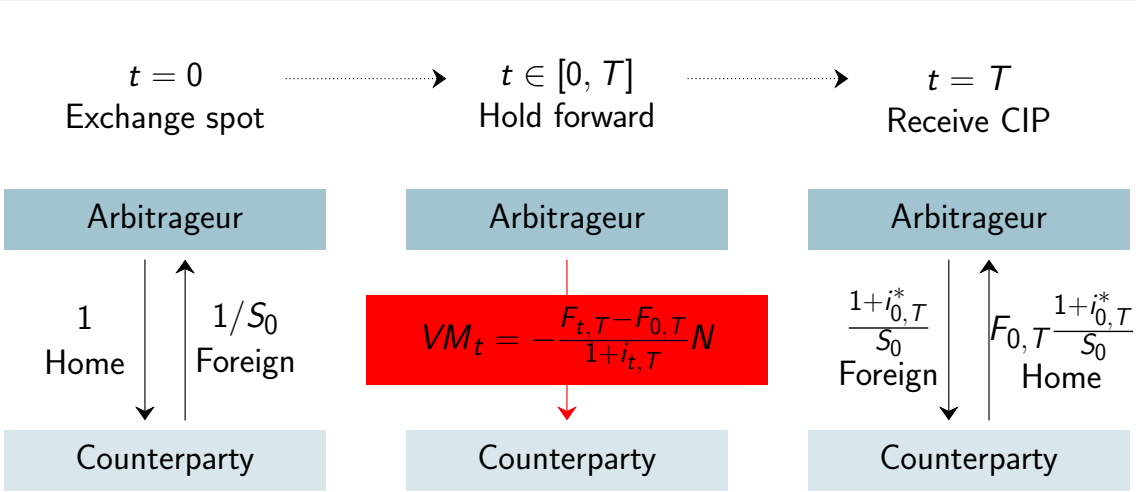
Duffie (2010); Mitchell et al. (2007); Siriwardane (2019); Siriwardane et al. (2023); Gabaix and Koijen (2020)...

- We use individual dealer P&L changes to map where internal capital constraints bind

CIP: Variation margin payments require (internal) liquidity, creating risk



CIP: Variation margin payments require (internal) liquidity, creating risk



How large are these variation margins? $\text{MaR}^{3M}_{95\%} = F^{-1}_{\mathcal{N}(0,1)}(\alpha)\sigma\sqrt{T} \simeq 12\%$!

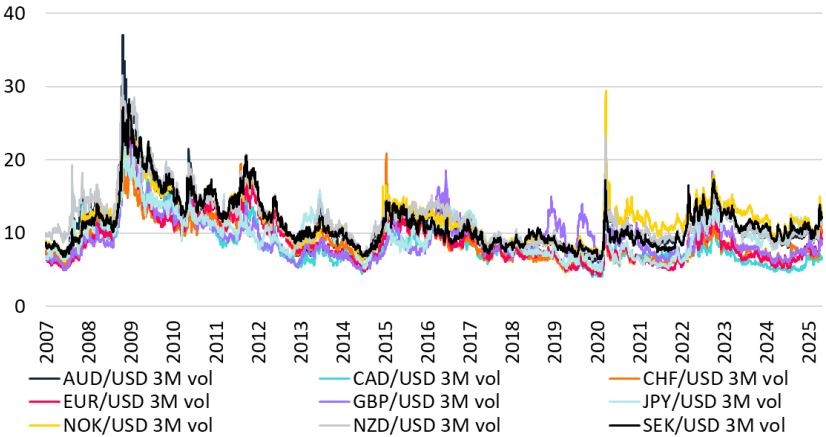


Figure: 3-month foreign exchange rate option-implied volatility. Source: LSEG.
Notes: The implied volatility is computed from at-the-money option quotes.

How large are these variation margins? At fair value through profit or loss

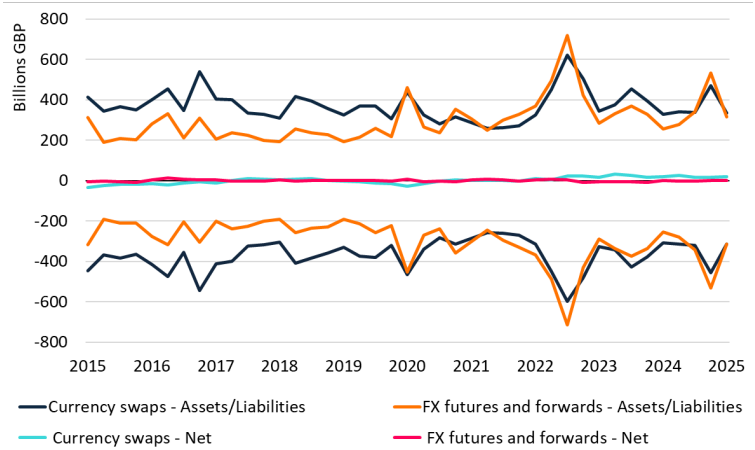


Figure: Quarterly amounts outstanding of UK resident banks' Foreign Exchange Swaps, Foreign Exchange Futures and Forwards - Liabilities and Assets (in billions GBP). Source: Bank of England.

How large are these variation margins? Assuming perfect netting

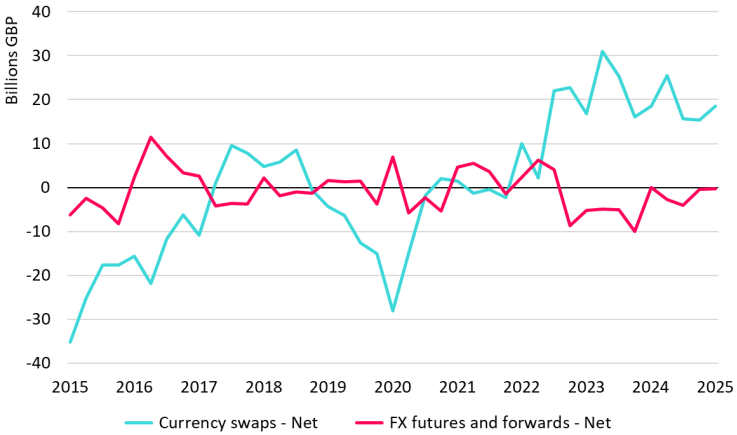


Figure: Quarterly amounts outstanding of UK resident banks' Foreign Exchange Swaps, Foreign Exchange Futures and Forwards - Liabilities and Assets (in billions GBP). Source: Bank of England.

A stylized model: Trade-off CIP deviations with the liquidity risk (1/3)

Key insight: Motivate Shleifer and Vishny (1997) with Bolton, Chen, and Wang (2011).

- 3 periods: $t \in \{0, 1, 2\}$, no discounting
- CIP arbitrage opportunities: $b = (b^c)_{c=1}^{\mathcal{C}} \neq 0$
- Initial period ($t = 0$): Choose $(x_0^c)_{c=1}^{\mathcal{C}}$
- Final payoff ($t = 2$): $\Pi_2^{\text{CIP}} = \sum_{c=1}^{\mathcal{C}} b^c x_0^c$
- Interim margin call ($t = 1$): uncertain, in state $\omega \in \Omega$, $M_1(\omega) = \sum_{c=1}^{\mathcal{C}} M_1^c(\omega) x_0^c$
- Liquidity shortfall: $S(\omega) = \min\left(M_1(\omega) + L_1(\omega), 0\right) = \left(M_1(\omega) + L_1(\omega)\right)^- \leq 0$
- Costly external funding: $\Phi : \mathbb{R}_- \to \mathbb{R}_+ \quad \Phi(0) = 0 \quad \Phi'(S) < 0 \quad \Phi''(S) > 0$

A stylized model: Trade-off CIP deviations with the liquidity risk (2/3)

The profit maximization of a representative bank's treasury is:

$$\max_{(x_0^c)_{c=1}^{\mathcal{C}}} \sum_{c=1}^{\mathcal{C}} b^c x_0^c - \mathbb{E}_0 \left[\Phi \left(\left(\sum_{c=1}^{\mathcal{C}} M_1^c(\omega) x_0^c + L_1(\omega) \right)^- \right) \right]$$

$$\forall c \in [\![1, \mathcal{C}]\!], b^c = \mathbb{E}_0 \left[M_1^c(\omega) \underbrace{\Phi' \left(S(\omega) \right) \mathbb{1}_{M_1(\omega)+L_1(\omega)\leq 0}}_{\Lambda(\omega)} \right]$$

- The SDF of the bank is the state-contingent shadow price of internal liquidity
- The CIP bases reflect the expected liquidity cost implied by the variation margins

A stylized model: Trade-off CIP deviations with the liquidity risk (3/3)

Asset pricing equation, with $M_1^c \sim \mathcal{N}(0, (\sigma^c)^2)$:

$$\forall c \in [\![1, \mathcal{C}]\!], b^c = \mathbb{E}_0 [\Lambda M_1^c] = \text{corr}(\Lambda, M_1^c) \cdot \sigma_\Lambda \cdot \sigma^c$$

- A currency with larger volatility of variation margins will have a larger basis
- A negative shock to the available liquidity L_1 will lead to larger bases
- A currency with variation margins that comove with the available liquidity L_1 will have a larger basis

$\Delta b_{c,t}$	(1)	(2)	(3)	(4)	(5)
$\Delta Spot_{c,t}$	0.1283				0.0231
	(0.1195)				(0.0901)
$\Delta i_{c,t}$		-0.0347**			-0.0417***
		(0.0124)			(0.0132)
$\Delta i_{USD,t}$			0.1102***		0.0905***
			(0.0265)		(0.0156)
$\Delta IVol_{c,t}$				-0.1245***	-0.0886***
				(0.0358)	(0.0285)
R-squared	0.0003	0.0019	0.0085	0.0067	0.0133
Observations	33,743	26,512	33,743	33,743	26,512

Table: Key drivers of changes in the value of foreign exchange derivatives and CIP deviations.
Notes: We regress the daily changes in the 3-month CIP deviation $\Delta b_{c,t}$ for currency c against the US dollar (computed as in Siriwardane et al. (2023)) on the daily log changes in the bilateral spot rate $\Delta Spot_{c,t}$, the 3-month OIS rates in currency c and in USD $\Delta i_{c,t}$ and $\Delta i_{USD,t}$, and the 3-month implied volatility of currency pair c/USD $\Delta IVol_{c,t}$, with currency pair fixed effects. Robust standard errors in parentheses, two-way clustered by currency and date. *** p<0.01, ** p<0.05, * p<0.1.

(Selected) Literature in International Macroeconomics and Finance

Imperfect financial intermediation is key (Itskhoki and Mukhin, 2021)

- Noise-trader demand shocks ψ_t + risk-averse financiers in a GE model
- can explain all major exchange rate puzzles (Meese-Rogoff, Backus-Smith, PPP, and UIP)
- UIP: $i_{0,1} - i^*_{0,1} - \mathbb{E}_0[s_1 - s_0] = \psi_0$... but CIP holds!

Intermediary AP theory can explain the limited risk bearing (Gabaix and Maggiori, 2015)

So maybe CIP really is not about risk? (Du, Tepper, and Verdelhan, 2018)

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Intermediary AP theory can explain the limited risk bearing (Gabaix and Maggiori, 2015)

- Only financiers can trade bonds internationally (Gamma model),
- and financiers can divert a share of intermediated funds $\Gamma = \gamma \ var(S)^{\alpha}$, with $\gamma, \alpha \geq 0$.

$$\max_{x_0} V_0 = \mathbb{E}_0 \left[(1 + i_{0,1}) - (1 + i^*_{0,1}) \frac{S_1}{S_0} \right] x_0, \qquad \text{subject to} \qquad V_0 \geq \Gamma x_0^2$$

$$x_0 = \tfrac{1}{\Gamma} \mathbb{E}_0 \left[(1 + i_{0,1}) - \left(1 + i^*_{0,1}\right) \tfrac{S_1}{S_0} \right] \underset{UIP}{\Longrightarrow} i_{0,1} - i^*_{0,1} - \mathbb{E}_0[s_1 - s_0] = \Gamma x_0 \ldots \text{ but CIP holds!}$$

So maybe CIP really is not about risk? (Du, Tepper, and Verdelhan, 2018)

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So maybe CIP really is not about risk? (Du, Tepper, and Verdelhan, 2018)

- The leverage ratio is a non-risk based measure
- LR = (CET1 + AT1) / exposure $\geq$ 3% (US: eSLR $\geq$ 5-6%), exposure $\underset{(\lambda)}{\propto} N$

$$\max_{(x_0^c)_{c=1}^{\mathcal{C}}} \sum_{c=1}^{\mathcal{C}} \left[\underbrace{(1+i_{0,1}) - \left(1+i_{0,1}^*\right) \frac{F_{0,1}}{S_0}}_{b^c} \right] x_0^c \qquad \underset{[\mu_0 \geq 0]}{\text{subject to}} \qquad \sum_{c=1}^{\mathcal{C}} \lambda \left| x_0^c \right| \leq K/0.03$$

$\forall c \in [\![1, \mathcal{C}]\!], |b^c| = \lambda \mu_0$, CIP bases are equalized: cannot explain the cross-section
(But CIP bases are a sufficient statistic for the stress of the banking sector)

CIP: The Revenge of the Currency Risk?

UIP and CIP are fundamentally linked through intermediaries and FX hedging demand

$$\underbrace{\left(i_{0,1} - i^*_{0,1}\right) - \mathbb{E}_0\left[\Delta s_1\right]}_{\text{UIP deviation}} \equiv \underbrace{f_{0,1} - \mathbb{E}_0[s_1]}_{\text{FX risk premium}} + \underbrace{\left(i_{0,1} - i^*_{0,1}\right) - \left(f_{0,1} - s_0\right)}_{\text{CIP deviation } b}$$

Variation margins make CIP look at lot like UIP: the original limits to arbitrage

- Daily, full, cash-settled VMs on derivatives' mark-to-market value (Basel III + risk mgmt)
- Key friction: internal funds are valuable (because external funds are costly)

(UK-)EMIR data: The Regulation

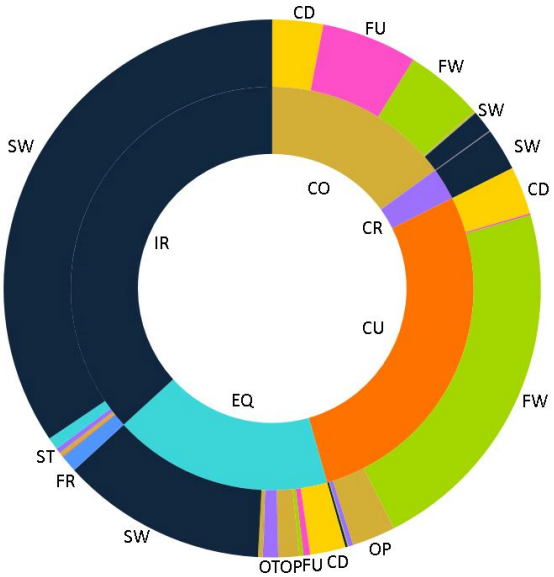
(United Kingdom) **European Market Infrastructure Regulation** (EMIR):

- Coverage: UK entities as (at least one) counterparty

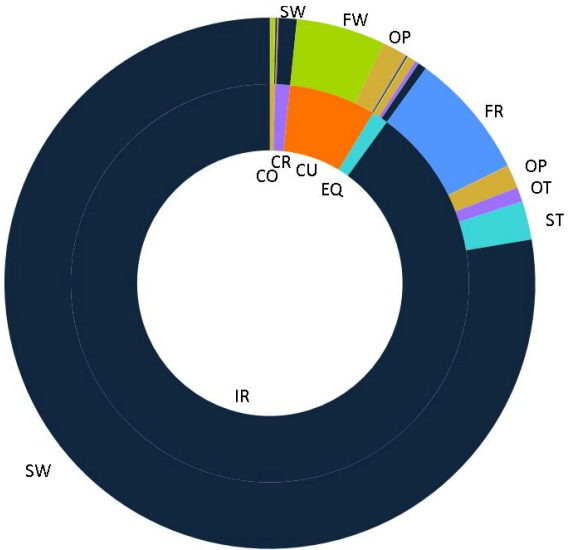
3 pillars: **Clearing, Reporting, Risk mitigation**

Data cleaning

Big data: ∼30 million open positions every day, with 129 fields by report for each counterparty.



(a) Distribution of open positions (24 million)



(b) Distribution of notional values (USD782 trillion)

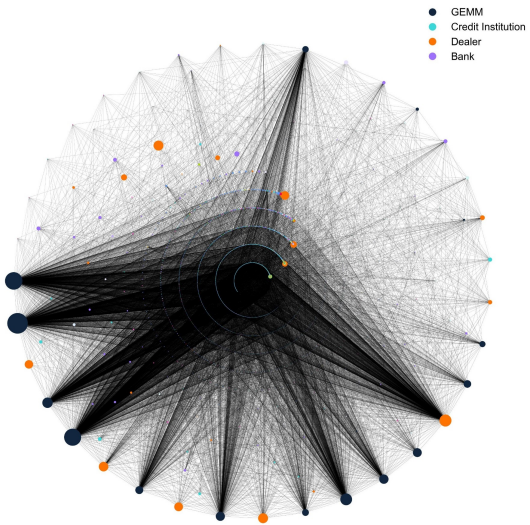
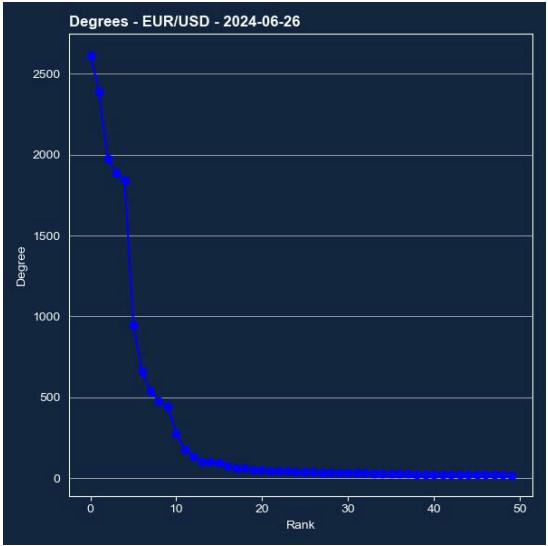
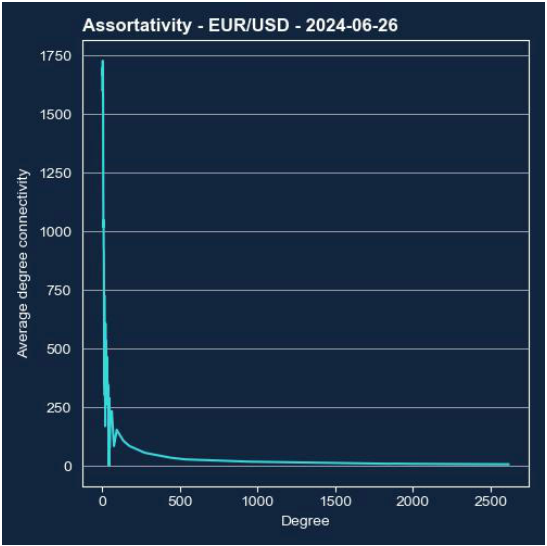


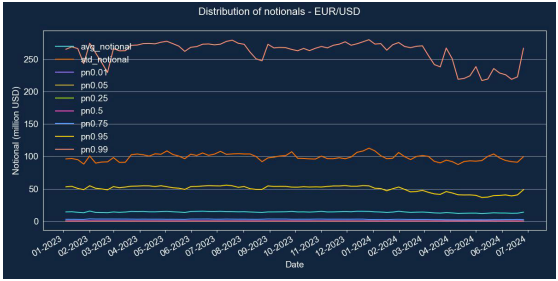
Figure: Network of outstanding derivative transactions in EUR/USD FX forwards.
Source: EMIR position data (stock) reported to the Bank of England (26 June 2024).



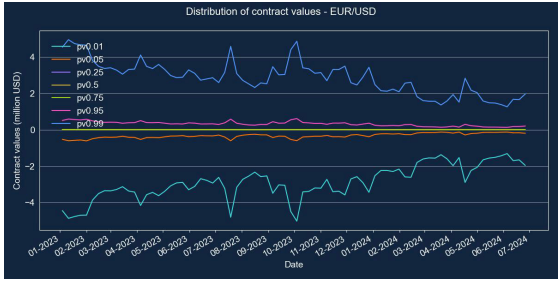
(a) Degrees



(b) Assortativity



(a) Distribution of notionals



(b) Distribution of contract values

Variation margins for dealer banks are very large

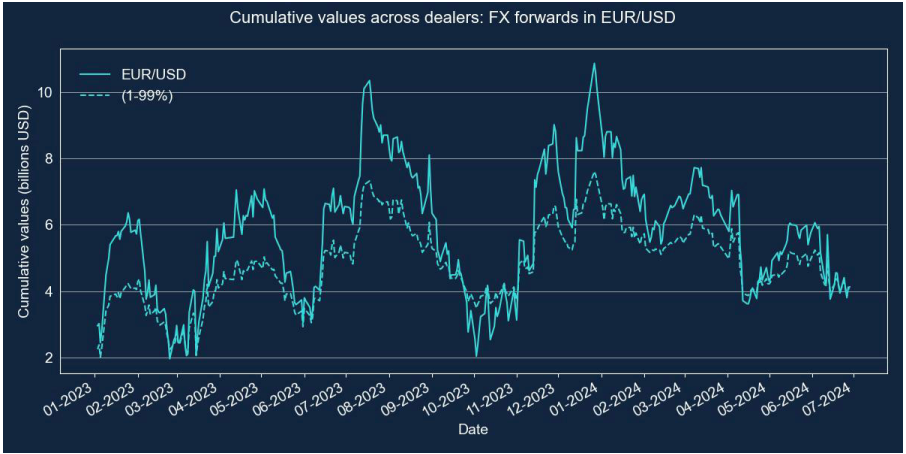


Figure: Cumulative net values of EUR/USD FX forwards for the main dealers in $\mathbb{D}$.
Sources: EMIR position data (stock) reported to the Bank of England (1 January 2023 - 30 June 2024).

Calculate CIP deviations at the contract-level

$$b_T = \left[\left(1 + \frac{i_{t,T}}{100} \frac{T-t}{DayC} \right) - \left(1 + \frac{i^*_{t,T}}{100} \frac{T-t}{DayC^*} \right) \frac{F_{0,T}}{S_0} \right] \times \frac{360}{T-t} \times 100$$

where:

- b_T is the contract level basis,
- $F_{0,T}$ is the contract-level forward rate,
- $i_{0,T}$ and $i^*_{0,T}$ zero coupon OIS curves from LSEG, interpolated with cubic splines,
- S_0 is the spot rate of the execution date,
- t is the spot settlement date, T the maturity date,
- we normalize the day count convention to the US Actual/360.

Results: Large dispersion across dealers and clients, consistent with OTC markets, $F_{0,T} > F^{CIP}_{0,T}$.

Main regression

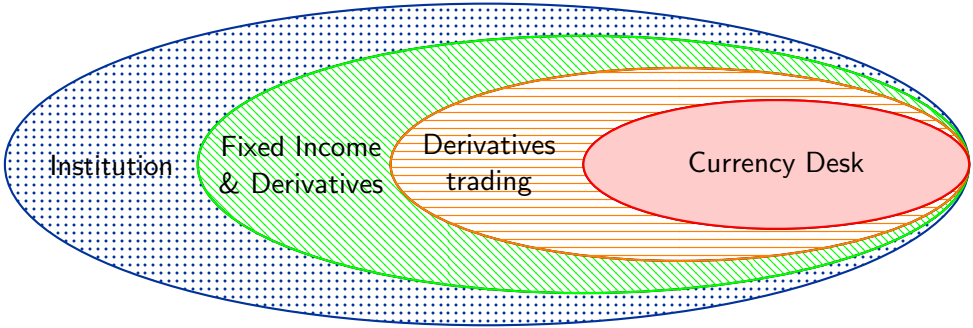


Figure: Partition of the balance-sheet of a major broker dealer into different levels.

$$\Delta b_{i,t} = \sum_{j \in \mathbb{J}} \beta_j \times \sum_{d(i) \in \mathbb{D}(i)} \omega_{d(i)} \Delta P\&L_{d(i),j,t-1} + controls_t + \epsilon_{i,t}$$

Main regression

$$\Delta b_{i,t} = \sum_{j \in \mathbb{J}} \beta_j \times \sum_{d(i) \in \mathbb{D}(i)} \omega_{d(i)} \Delta P\&L_{d(i),j,t-1} + controls_t + \epsilon_{i,t}$$

where:

- $b_{i,t}$ is the basis swap at time t in currency pair i,
- $\mathbb{J}$ is the partition of the dealers' balance sheets into the foreign exchange derivatives trading desk, all the other derivatives, all the other fixed income activities, and the rest of the institution,
- $\mathbb{D}(i)$ is the set of dealers in currency pair i,
- $\omega_{d(i)}$ is the weight of dealer $d(i)$ in currency pair i (degree / share of total notionals)
- $P\&L_{d(i),j,t}$ is the profit and loss statement of dealer $d(i)$ at level j at time t.

Policy implications

- Variation margins transform credit risk into liquidity risk, with a potential trade-off.
  - Granularity + Different marginal values of cash = **Not neutral** for the financial system
  - Liquidity preparedness for margin calls is key (BCBS-CPMI-IOSCO and FSB work)
- VM liquidity risk is sizable even in the most liquid OTC derivative market (billions USD)
  $\Longrightarrow$ Less liquid markets with the same structure should be worse.
- It is not only about the non-risk-based leverage ratio or RWA
  $\Longrightarrow$ LCR/NSFR and liquidity risk management also govern arbitrage capacity.
- The granularity of dominant intermediaries can be a source of systemic risk
  $\Longrightarrow$ Not only about initial margins, monitor VM concentration with EMIR data.

Conclusion

- **Hypothesis**: CIP bases could also be the price of internal-liquidity risk created by VMs
- **Mechanism**: Intermediary SDF is the shadow price of internal liquidity $b^c = \mathbb{E}[\Lambda M_1^c]$
- **Data**: Document the market structure of FX forwards (UK-EMIR, contract level)
  - Large, granular dealers intermediate the OTC, bilateral market $\implies$ non-neutrality
  - Large and changing variation margins on outstanding derivatives $\implies$ liquidity risk
  - Significant dispersion in forward rates across dealers and clients $\implies$ OTC frictions
- **Ongoing**: Empirical identification, with a GIV approach
- **New channel**: the binding friction is internal liquidity, not only capital, which can explain the cross-section of CIP deviations

  $\implies$ Bridge the gap between the intermediary asset pricing theory and empirics
- **Next:** Part of a larger agenda to use new financial datasets to answer macroeconomic questions

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The Leverage Ratio for a CIP trade: What Basel III says (with full funding)

Current Exposure Method (before 2022)

$$\text{CEM} = \underbrace{N}_{\text{funding}} + \underbrace{\max(V - VM,\, 0)}_{\substack{\text{RC=0} \\ \text{(full daily cash VM)}}} + \underbrace{1\%}_{\text{FX add-on}} \times \underbrace{N}_{\text{notional}} = 1.01\, N$$

Standardized Approach (SA-CCR, after 2022)

$$\text{SA-CCR} = \underbrace{N}_{\text{funding}} + \underbrace{1.4}_{\alpha} \left(\underbrace{\max(V - VM, 0)}_{\substack{\text{RC=0} \\ \text{(full daily cash VM)}}} + 1 \cdot \underbrace{4\%}_{\text{SF}_{\text{FX}}} \cdot \underbrace{N}_{\text{notional}} \cdot \underbrace{\frac{3}{2}\sqrt{\frac{MPOR}{1\text{year}}}}_{\text{maturity factor}} \right) = 1.0168\, N$$

But the bank can also marginally reduce non-USD deposits while increasing USD funding to earn the CIP, so that only the FX forward increases the balance-sheet exposure ([1-2]% $\times$ N)

Funds Transfer Pricing (FTP)

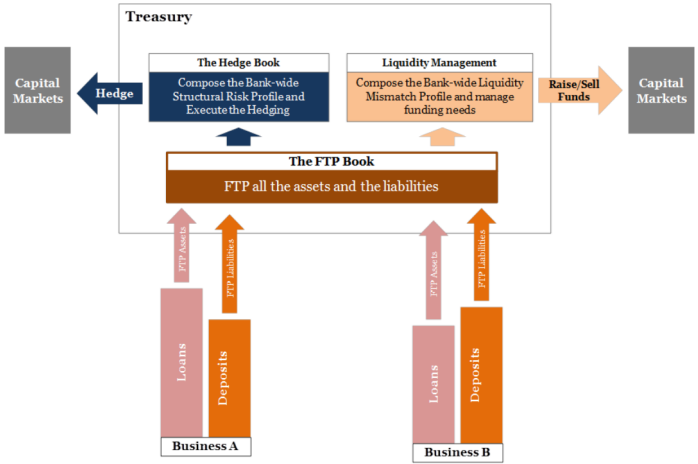


Figure: Assets and liabilities transferred into a FTP book (Tumasyan, 2012)

(UK-)EMIR data: Reporting

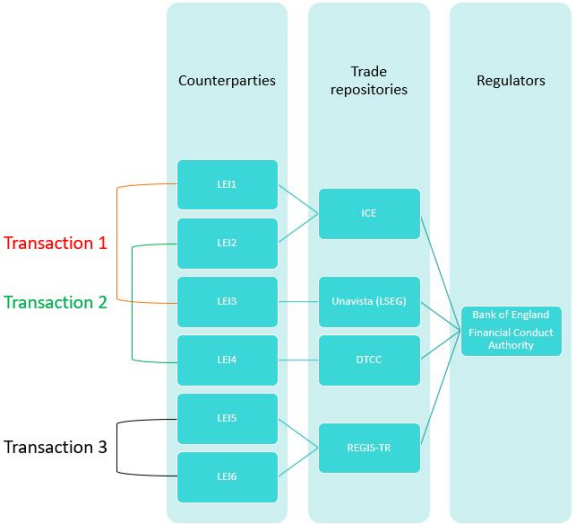


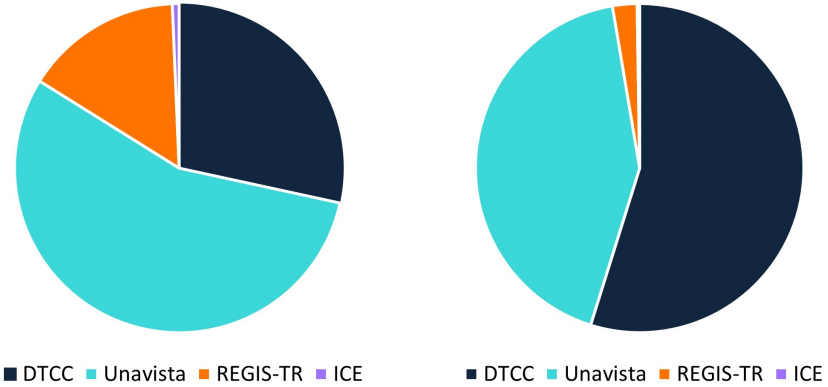
Illustration of the reporting process:

- 2 counterparties enter a derivative contract (suppose both UK-based)
- Each counterparty chooses a TR
- Regulators receive data from TRs

Double reporting:

+ Can compare the 2 sides
– Risk of double counting

(UK-)EMIR data: Trade repositories

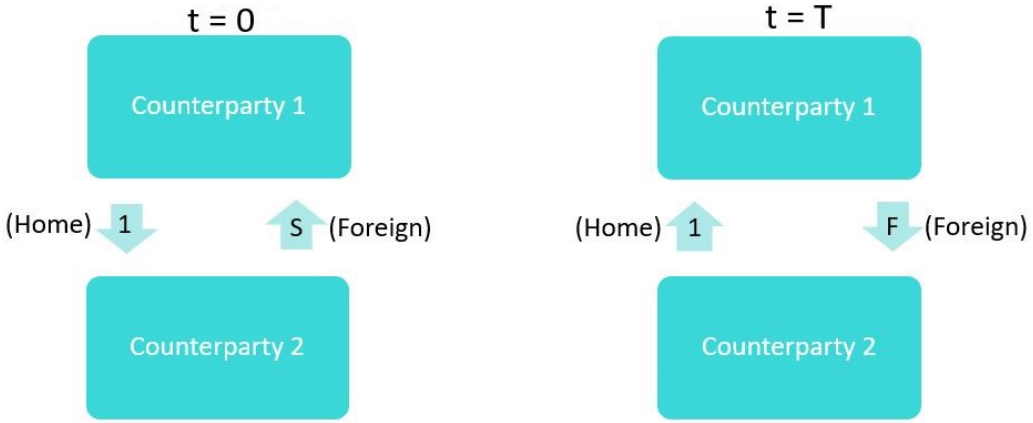


(a) Distribution of open positions

(b) Distribution of notional values

Figure: The market for reporting OTC FX derivatives

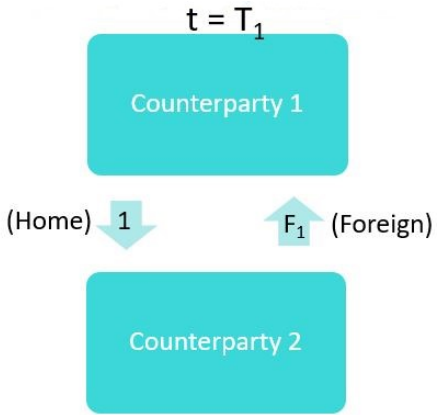
FX swaps as FX forwards? (1)

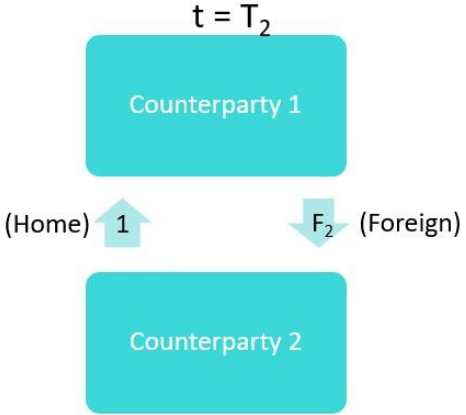


EMIR is about derivatives, not spot transactions.

- Only the forward leg is a derivative, hence the transaction reported as an FX forward.

FX swaps as FX forwards? (2)





This FX swap should be reported as a swap, given that it has two forward legs.

- Market convention is to report the two FX forwards independently.

(UK-)EMIR data: Comparison to the BIS data

	(UK-)EMIR	**BIS Triennial Survey**	**BIS OTC derivatives**
Coverage	All derivatives	FX spot, FXD, IRD	OTC derivatives
Scope	All UK-based entities	"Reporting dealers" $>$ 50 jurisdictions	Dealer banks 12 jurisdictions
Stock or Flow	Stock and Flow	Flow	Stock
Level	Individual	Jurisdiction-sector	Jurisdiction-sector
Consolidation	None	None	Worldwide excluding intragroup
Frequency	Daily	Every 3 years	Every 6 months
Access	Confidential Some public aggregates	Public (jurisdiction-sector)	Public (sector)

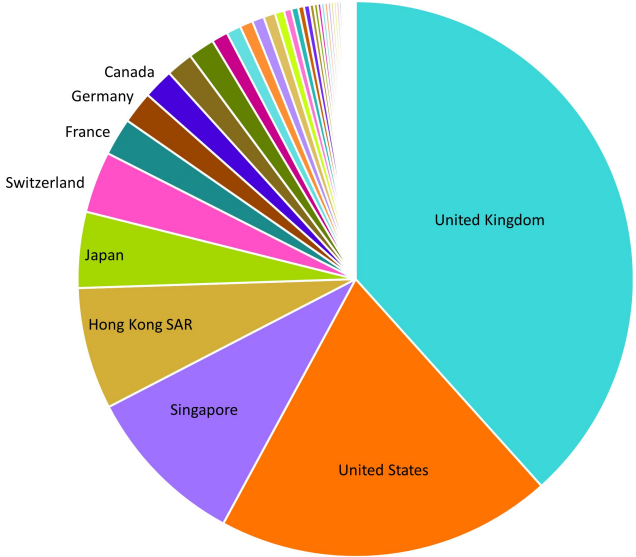


Figure: **Turnover by country** (Outright forwards, FX swaps and currency swaps, BIS, 2022)

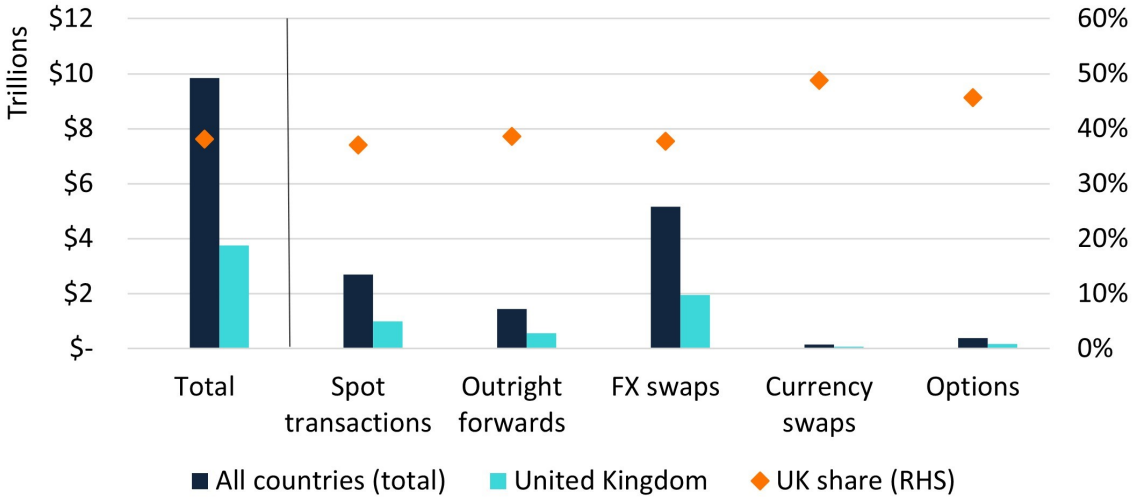


Figure: **Daily average turnover** (OTC foreign exchange "net-gross", BIS, 2022)

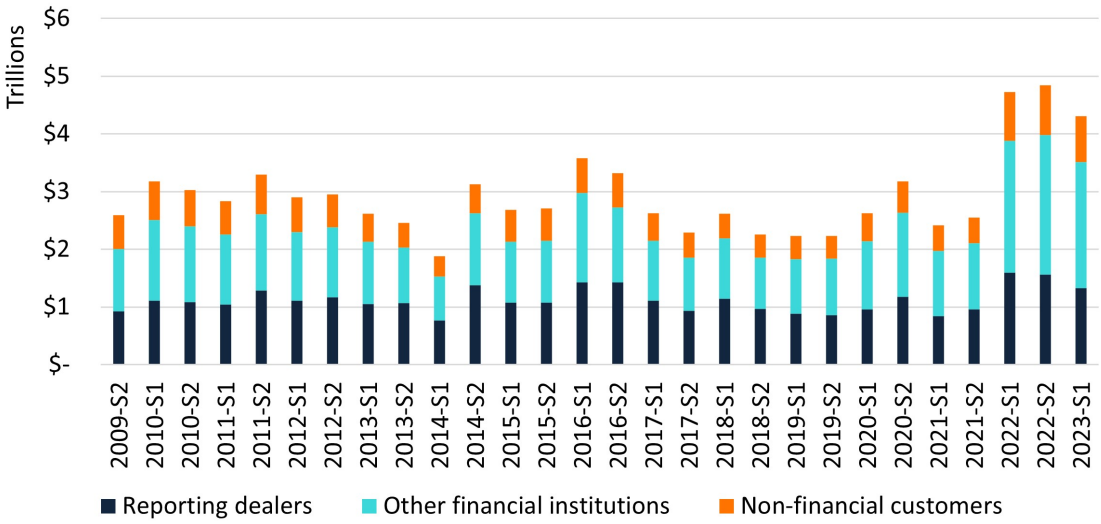


Figure: **FX derivatives gross market value** (forwards, FX swaps and currency swaps, BIS, S1-2023)

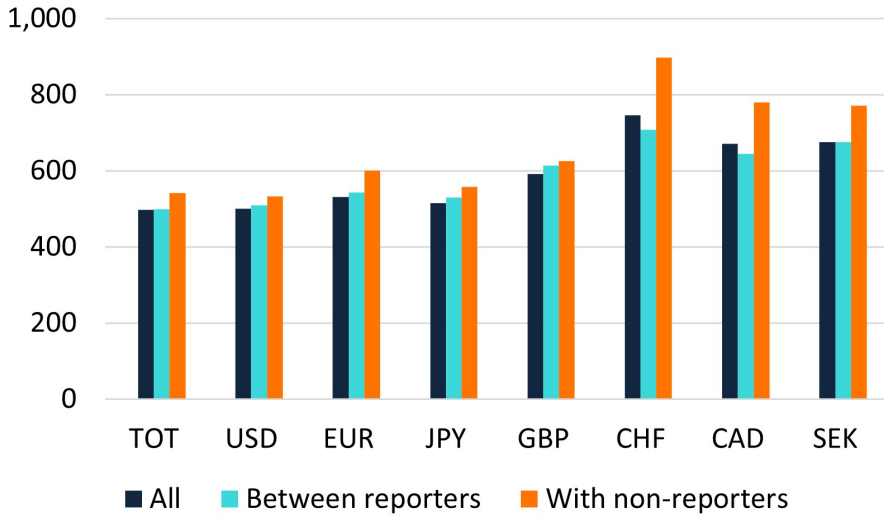


Figure: **HHI** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)

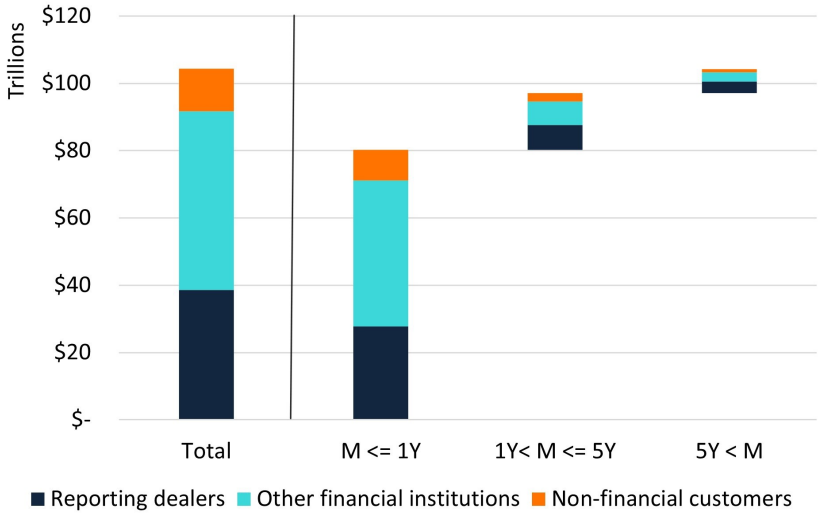


Figure: **Notionals outstanding** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)

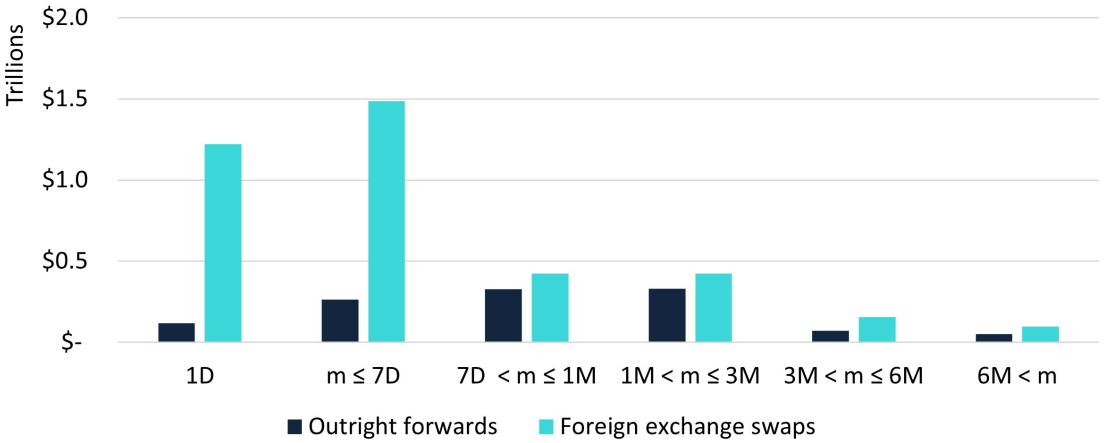


Figure: **Turnover by maturity** (Outright forwards, FX swaps and currency swaps, BIS, 2022)

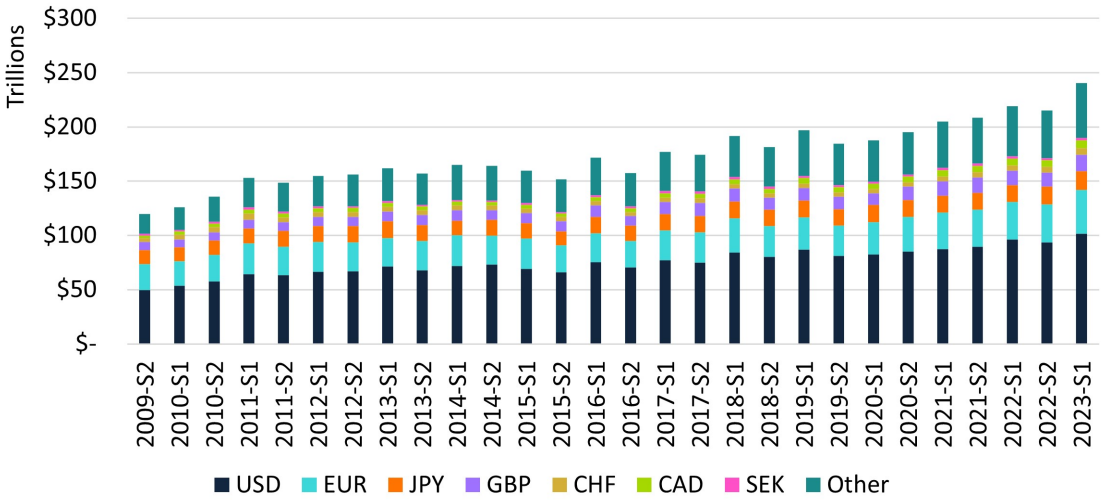


Figure: **Outstanding by currency** (Outright forwards, forex swaps and currency swaps, BIS, S1-2023)